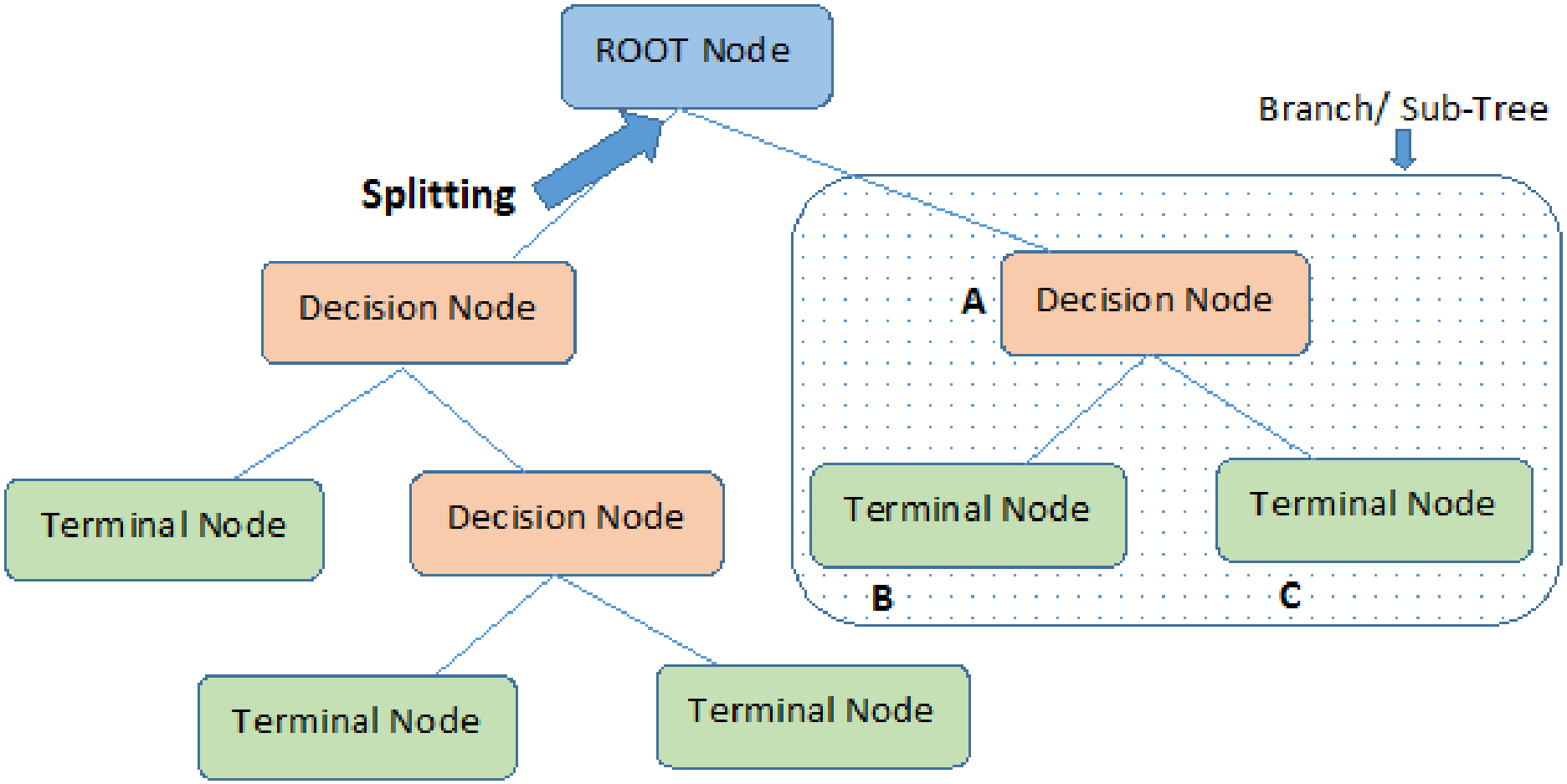
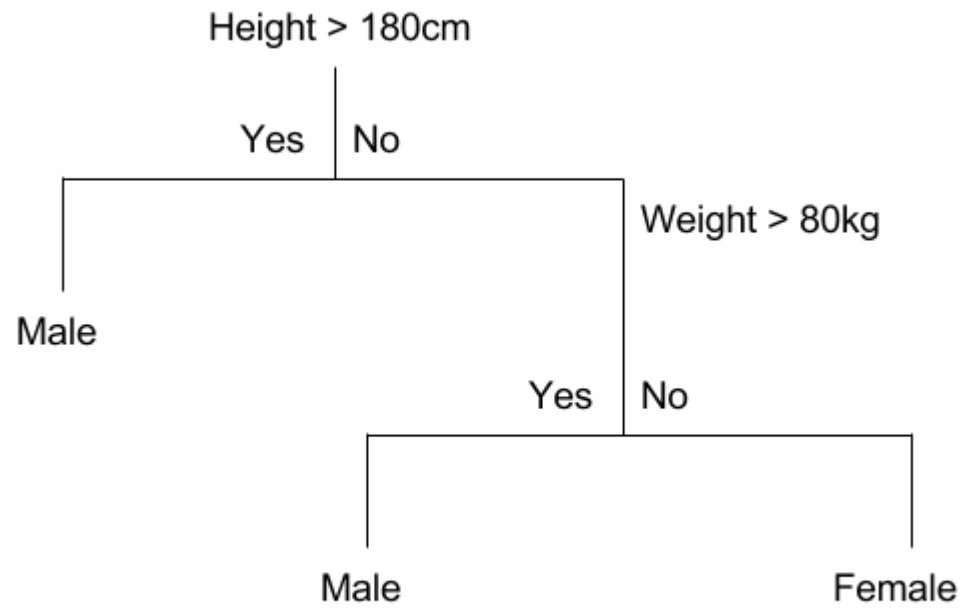
# Decision Tree


ROOT Node
Branch/ Sub-Tree
Splitting
Decision Node
A
Decision Node
Terminal Node
Decision Node
Terminal Node
Terminal Node
B
C
Terminal Node
Terminal Node

The decision tree is one of the most important machine learning algorithms.
It is used for both classification and regression problems.
Here, we will go through the classification part.

**What is a decision tree?**

A decision tree is a classification and prediction tool having a tree-like structure,
where each internal node denotes a test on an attribute,
each branch represents an outcome of the test, and
each leaf node (terminal node) holds a class label.




Above we have a small decision tree.
An important advantage of the decision tree is that it is highly interpretable.
Here If Height > 180cm or if height < 180cm and weight > 80kg person is male.
Otherwise female.

important terms related to decision trees.

**Entropy**

In machine learning, entropy is a measure of the randomness in the information being processed.
The higher the entropy, the harder it is to draw any conclusions from that information.

$$\mathrm{H}(X) = -\sum_{i=1}^{n} \mathrm{P}(x_i) \log_b \mathrm{P}(x_i)$$

## Information Gain

Information gain can be defined as the amount of information gained about a random variable or signal from observing another random variable.
It can be considered as the difference between the entropy of parent node and weighted average entropy of child nodes.
Information gain is the amount of knowledge acquired during a certain decision, whereas entropy is a measure of uncertainty or unpredictability.
Information gain is the reduction in entropy by transforming the dataset.
IG is a statistical property that measures how well a given attribute separates the training examples according to their target classification. IG value is always between 0 and 1.

$$IG(S,A) = H(S) - H(S,A)$$

Alternatively,

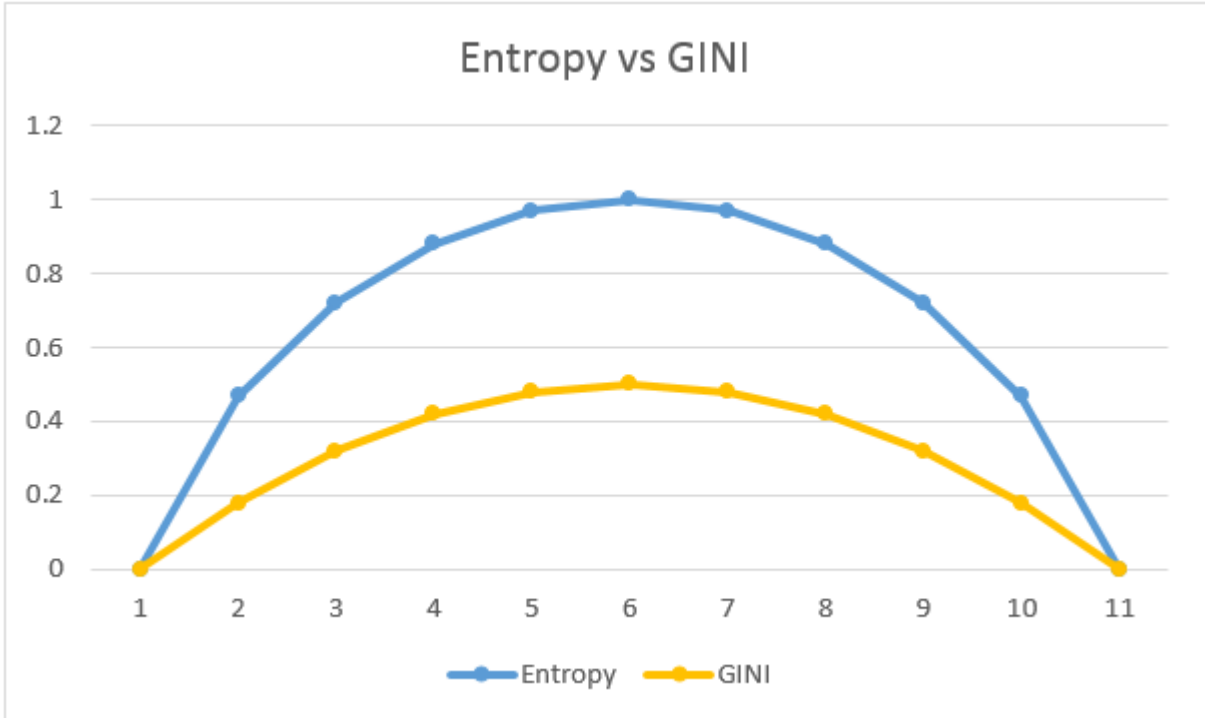
$$IG(S,A) = H(S) - \sum_{i=0}^{n} P(x) * H(x)$$

## Gini Impurity

Gini impurity is a measure of how often a randomly chosen element from the set would be incorrectly labeled if it was randomly labeled according to the distribution of labels in the subset.
Gini index is the measure of impurity or entropy in the values of a dataset.
Gini index is the probability of misclassification when an instance is randomly classified.

$$Gini(E) = 1 - \sum_{j=1}^{c} p_j^2$$

Gini impurity is **lower bounded by 0**, with 0 occurring if the data set contains only one class.


Entropy vs GINI
1.2
1
0.8
0.6
0.4
0.2
0
1
2
3
4
5
6
7
8
9
10
11
Entropy
GINI

There are many algorithms there to build a decision tree. They are

**1.CART** (Classification and Regression Trees) — This makes use of Gini impurity as the metric.
**2.ID3** (Iterative Dichotomiser 3) — This uses entropy and information gain as metric.

**Classification using the ID3 algorithm**

Consider whether a dataset based on which we will determine whether to play football or not.

| Outlook | Temperature | Humidity | Wind | Played football(yes/no) |
|---|---|---|---|---|
| Sunny | Hot | High | Weak | No |
| Sunny | Hot | High | Strong | No |
| Overcast | Hot | High | Weak | Yes |
| Rain | Mild | High | Weak | Yes |
| Rain | Cool | Normal | Weak | Yes |
| Rain | Cool | Normal | Strong | No |
| Overcast | Cool | Normal | Strong | Yes |
| Sunny | Mild | High | Weak | No |
| Sunny | Cool | Normal | Weak | Yes |
| Rain | Mild | Normal | Weak | Yes |
| Sunny | Mild | Normal | Strong | Yes |
| Overcast | Mild | High | Strong | Yes |
| Overcast | Hot | Normal | Weak | Yes |
| Rain | Mild | High | Strong | No |

In the dataset there are four independent variables to determine the dependent variable.
The independent variables are Outlook, Temperature, Humidity, and Wind.
The dependent variable is whether to play football or not.

As the first step, we have to find the parent node for our decision tree. For that follow the steps:

***Find the entropy of the class variable.***

$E(S) = -[(9/14)\log(9/14) + (5/14)\log(5/14)] = 0.94$

**note**: Here typically we will take log to base 2.
Here total there are 14 yes/no.
Out of which 9 yes and 5 no.
Based on it we calculated probability above.

From the above data for outlook we can arrive at the following table easily

| | | play | | |
|---|---|---|---|---|
| | | yes | no | total |
| | sunny | 3 | 2 | 5 |
| Outlook | overcast | 4 | 0 | 4 |
| | rainy | 2 | 3 | 5 |
| | | | | 14 |

***Now we have to calculate average weighted entropy***. ie, we have found the total of weights of each feature multiplied by probabilities.

E(S, outlook) = (5/14)*E(3,2) + (4/14)*E(4,0) + (5/14)*E(2,3) = (5/14)(-(3/5)log(3/5)-(2/5)log(2/5))+ (4/14)(0) + (5/14)((2/5)log(2/5)-(3/5)log(3/5)) = 0.693

***The next step is to find the information gain***.
It is the difference between parent entropy and average weighted entropy we found above.

IG(S, outlook) = 0.94 - 0.693 = 0.247

Similarly find Information gain for Temperature, Humidity, and Windy.

IG(S, Temperature) = 0.940 - 0.911 = 0.029

IG(S, Humidity) = 0.940 - 0.788 = 0.152

IG(S, Windy) = 0.940 - 0.8932 = 0.048

***Now select the feature having the largest entropy gain***.
Here it is Outlook.
So it forms the first node(root node) of our decision tree.

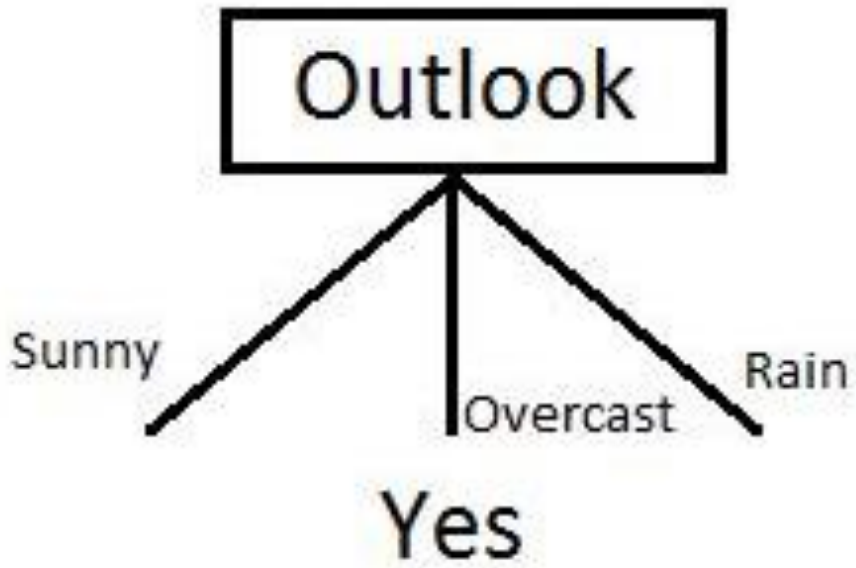
Now our data look as follows

| Outlook | Temperature | Humidity | Wind | Played football(yes/no) |
|---|---|---|---|---|
| Sunny | Hot | High | Weak | No |
| Sunny | Hot | High | Strong | No |
| Sunny | Mild | High | Weak | No |
| Sunny | Cool | Normal | Weak | Yes |
| Sunny | Mild | Normal | Strong | Yes |

| Outlook | Temperature | Humidity | Wind | Played football(yes/no) |
|---|---|---|---|---|
| Overcast | Hot | High | Weak | Yes |
| Overcast | Cool | Normal | Strong | Yes |
| Overcast | Mild | High | Strong | Yes |
| Overcast | Hot | Normal | Weak | Yes |

| Outlook | Temperature | Humidity | Wind | Played football(yes/no) |
|---|---|---|---|---|
| Rain | Mild | High | Weak | Yes |
| Rain | Cool | Normal | Weak | Yes |
| Rain | Cool | Normal | Strong | No |
| Rain | Mild | Normal | Weak | Yes |
| Rain | Mild | High | Strong | No |

Since overcast contains only examples of class ‘Yes’ we can set it as yes.
That means If outlook is overcast football will be played.
Now our decision tree looks as follows.


Outlook
Sunny
Overcast
Rain
Yes

The next step is to find the next node in our decision tree.
Now we will find one under sunny.
We have to determine which of the following Temperature, Humidity or Wind has higher information gain.

| Outlook | Temperature | Humidity | Wind | Played football(yes/no) |
|---|---|---|---|---|
| Sunny | Hot | High | Weak | No |
| Sunny | Hot | High | Strong | No |
| Sunny | Mild | High | Weak | No |
| Sunny | Cool | Normal | Weak | Yes |
| Sunny | Mild | Normal | Strong | Yes |

Calculate parent entropy E(sunny)

E(sunny) = (-(3/5)log(3/5)-(2/5)log(2/5)) = 0.971.

Now Calculate the information gain of Temperature. IG(sunny, Temperature)

| | | play | | |
|---|---|---|---|---|
| | | yes | no | total |
| | hot | 0 | 2 | 2 |
| Temperature | cool | 1 | 1 | 2 |
| | mild | 1 | 0 | 1 |
| | | | | 5 |

E(sunny, Temperature) = (2/5)*E(0,2) + (2/5)*E(1,1) + (1/5)*E(1,0)=2/5=0.4

Now calculate information gain.

IG(sunny, Temperature) = 0.971–0.4 =0.571

Similarly we get

IG(sunny, Humidity) = 0.971

IG(sunny, Windy) = 0.020

Here IG(sunny, Humidity) is the largest value.
So Humidity is the node that comes under sunny.

| | play | |
|---|---|---|
| Humidity | yes | no |
| high | 0 | 3 |
| normal | 2 | 0 |

For humidity from the above table, we can say that play will occur if humidity is normal and will not occur if it is high.
Similarly, find the nodes under rainy.

***Note: A branch with entropy more than 0 needs further splitting.***

Finally, our decision tree will look as below:

